

Lecture 6:

Probability Theory Basics



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University of California, Berkeley

**Numerical Methods for
Computational Science**

MSSE 273, 3 Units

Course Map

Week 1: Introduction to Scientific Computing and Python Libraries

Week 2: Linear Algebra Fundamentals

Week 3: Vector Calculus

Week 4: Numerical Differentiation and Integration

Week 5: Solving Nonlinear Equations

Week 6: Probability Theory Basics

Week 7: Random Variables and Distributions

Week 8: Statistics for Data Science

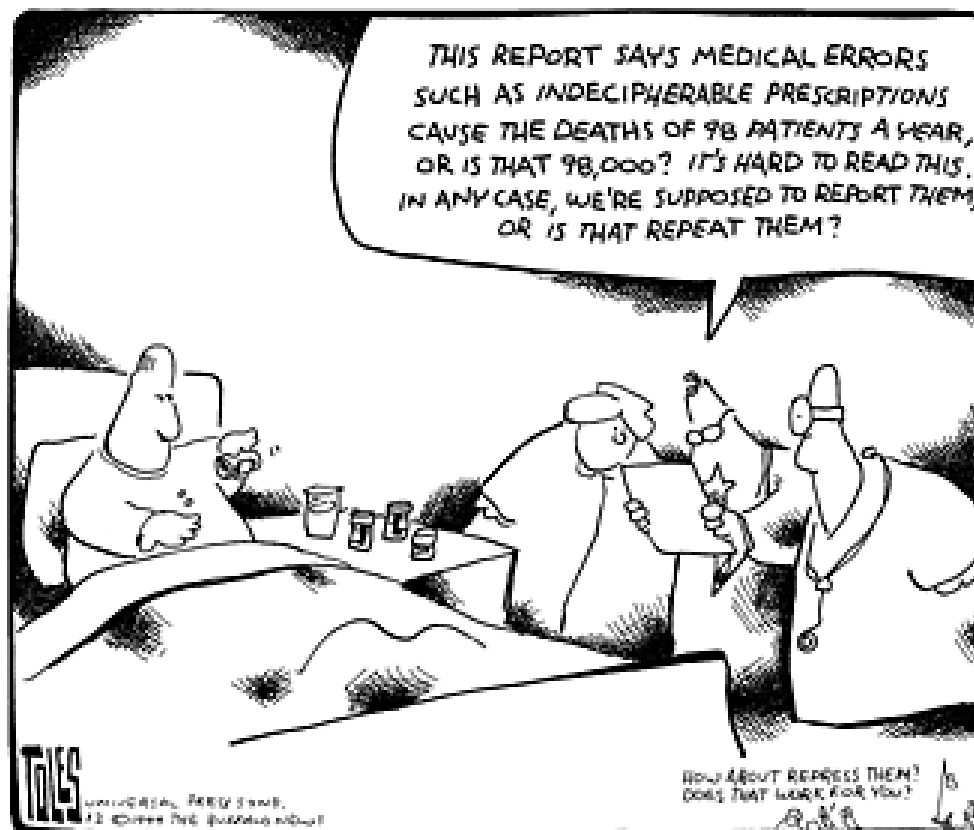
Week 9: Eigenvalues and Eigenvectors

Week 10: Simulation and Monte Carlo Method

Week 11: Data Fitting and Regression

Week 12: Optimization Techniques

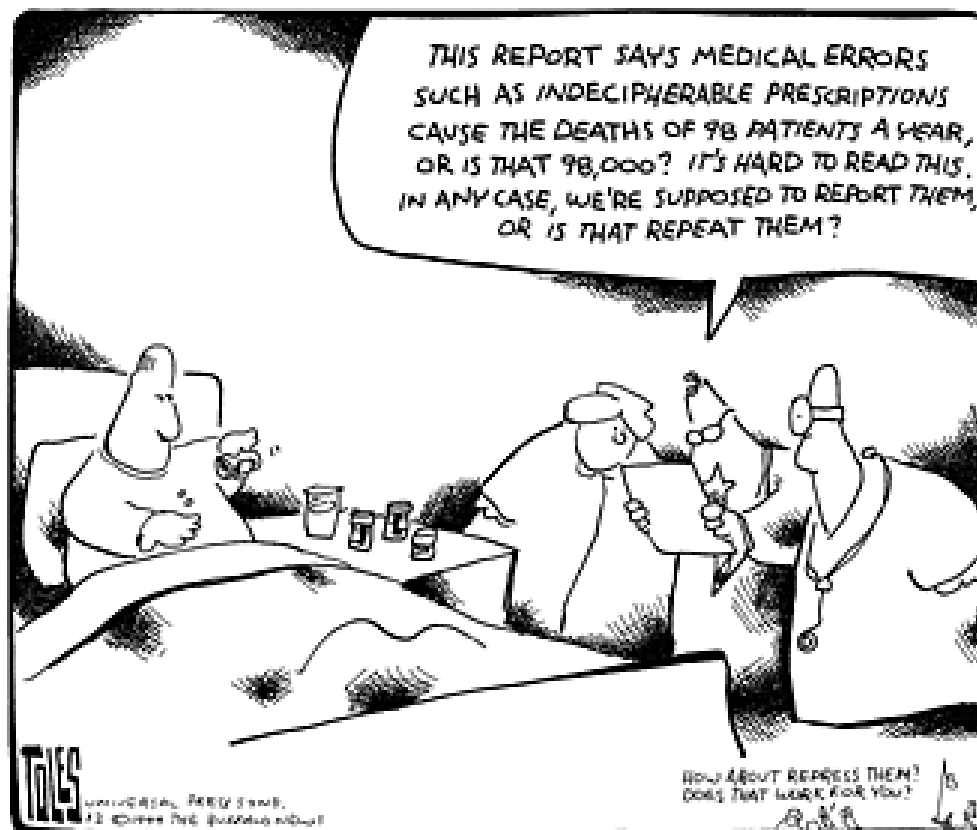
Week 13: Machine Learning Fundamentals



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Outline

- Axioms of Probability
- Conditional Probabilities and Bayes Theorem
- Information and Entropy



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Outline

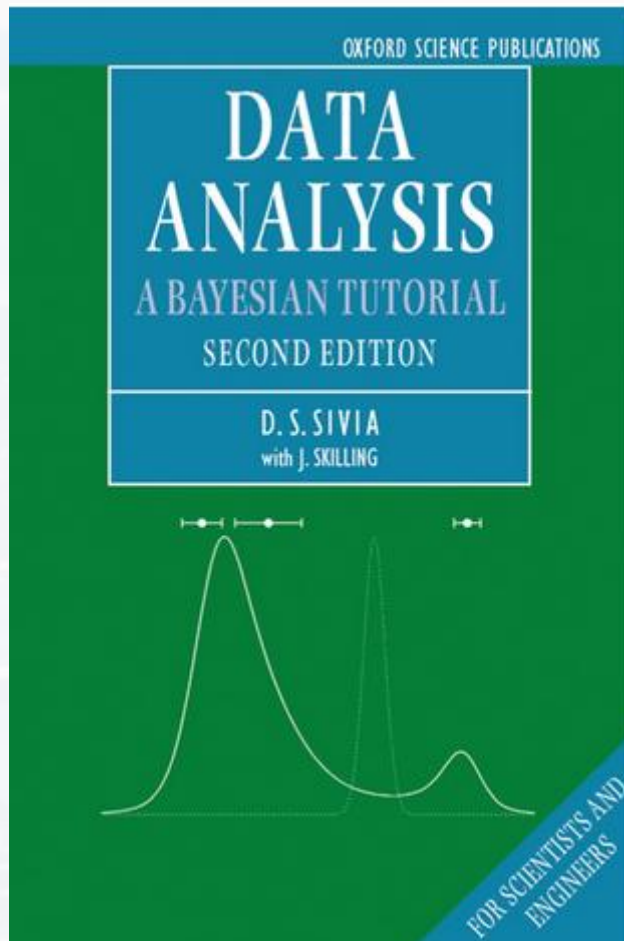
- Axioms of Probability

- Conditional Probabilities and
Bayes Theorem

- Information and Entropy



here: **heuristic explanation** → more mathematical rigorous: see **Cox's theorem**



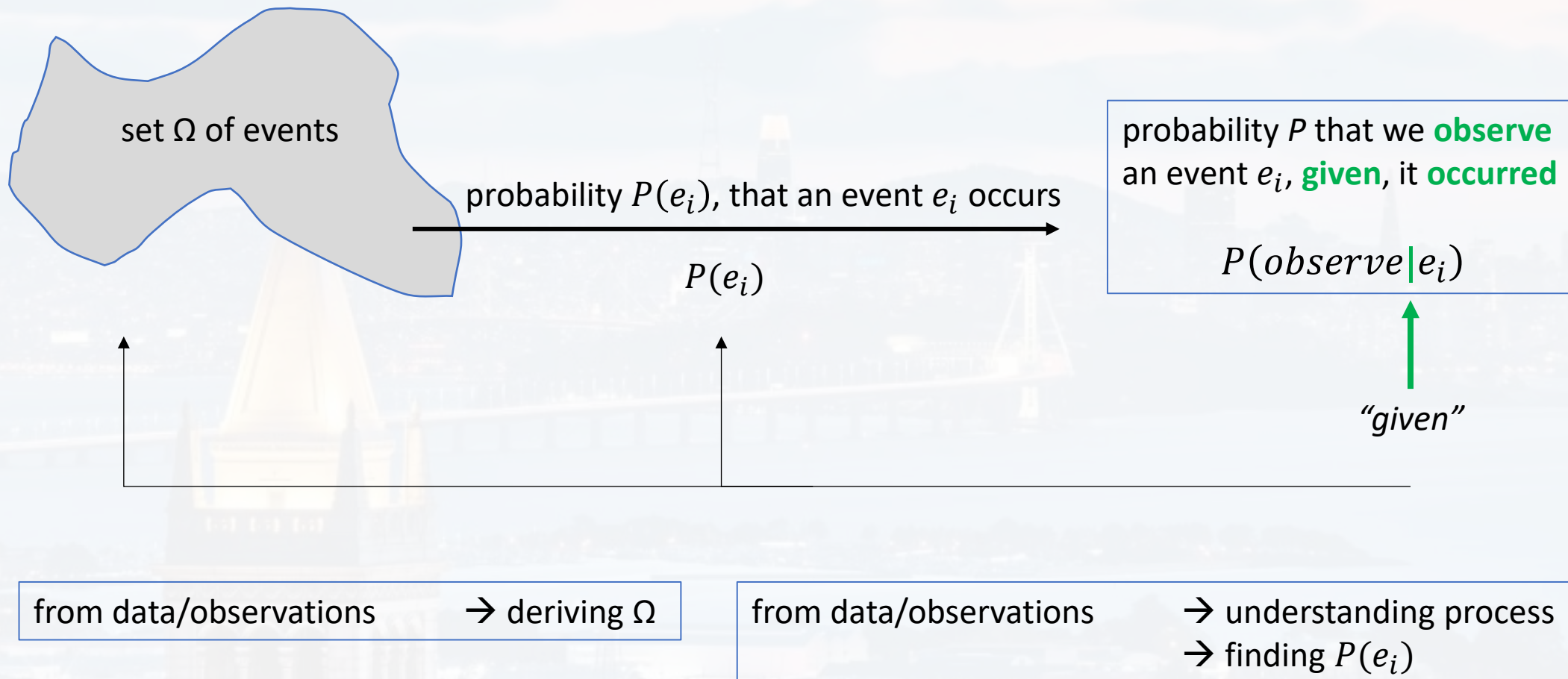
D. S. Sivia: *"Data Analysis"*

Bayesian Statistics

Appendix B: Cox's Theorem

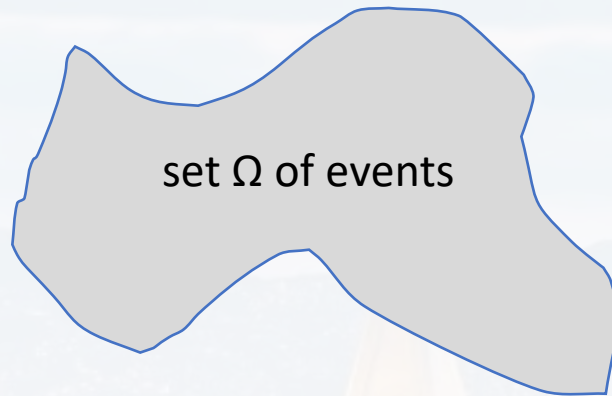


here: **heuristic explanation** → more mathematical rigorous: see **Cox's theorem**





here: **heuristic explanation** → more mathematical rigorous: see **Cox's theorem**



from data/observations

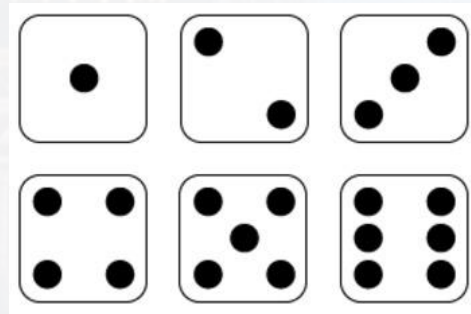
→ deriving Ω

1 2 2 1 2 2 1 2 1 1 2 1 1 2

observations

Is the observation 3 just rare (and that's why we haven't observed it), or is $3 \notin \Omega$?

events can be **discrete**

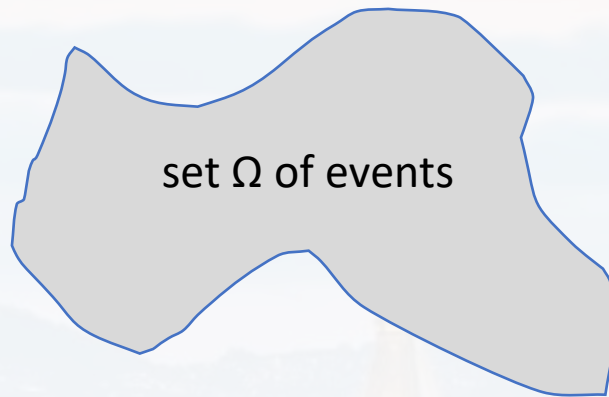


or **continuous**

- car speed measured in a speed trap
- a person's weight, etc



$P(e_i)$, that an event e_i occurs



1st axiom:

$P(e_i)$ is a **non-negative, real number**

2nd axiom:

the probability that at **least one** of the events in the entire sample space will occur is **1**
if events are **collectively exhaustive**

from 1st and 2nd: $P(e_i) = [0, 1]$ for any e_i

If events are **mutually exclusive**:

3rd axiom:

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

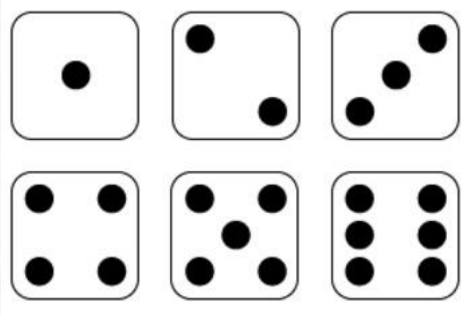
$\bigcup_{i=1}^{\infty} e_i$ means e_1 **or** e_2 **or** e_{∞}



$P(e_i)$, that an event e_i occurs

If events are **mutually exclusive**: 3rd axiom: $P(\bigcup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$

$\bigcup_{i=1}^{\infty} e_i$ means e_1 **or** e_2 **or** e_{∞}



The probability that we roll a **4 or a 6** equals...

$$P\left(e_4 \bigcup e_6\right) =$$

...the **probability** that we roll a **4 plus** the **probability** that we roll a **6**

$$P(e_4) + P(e_6)$$

“or” equals addition!



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

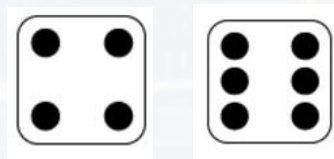
$P(e_i)$, that an event e_i occurs

“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

two dice: The probability that we roll a **4 and a 6** equals...

$$P\left(e_4 \cap e_6\right) =$$



...the **probability** that we roll a **4 times** the **probability** that we roll a **6**

$$P(e_4)P(e_6)$$



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

$P(e_i)$, that an event e_i occurs

“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

$\cup$: “union”

$\cap$: “intersection”

Be careful if events are not mutually exclusive (like a **set of events** or a **sequence of events**)

two light bulbs A and B:

What is the probability
that A or B is turned on?



A



B



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

$P(e_i)$, that an event e_i occurs

“and” equals multiplication!

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$\cup$: “union”

$\cap$: “intersection”

Be careful if events are not mutually exclusive (like a **set of events** or a **sequence of events**)

two light bulbs A and B:



A



B

What is the probability
that A **or** B is turned on?

It depends on how you define
“or”



two light bulbs A and B: **EX**clusive **OR**

What is the probability that A **or** B is turned on?



A



B



A



B



A



B



A



B



$$P(A) + P(B)$$

U: "union"

∩: "intersection"

$$P(A)P(B)$$

$$P\left(A \cup B\right) = P(A) + P(B) - 2P\left(A \cap B\right)$$

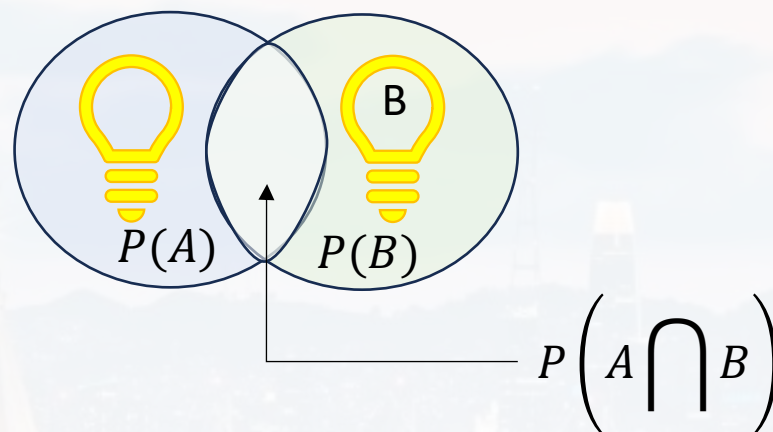
$$= P(A) + P(B) - 2P(A)P(B)$$



two light bulbs A and B: **EX**clusive **OR**

What is the probability
that A **or** B is turned on?

Venn diagram:



Note: Be careful with Venn diagrams!
They are not a mathematical proof!

$\cup$: “union”

$\cap$: “intersection”

size of **blue** area: $P(A)$

size of **green** area: $P(B)$

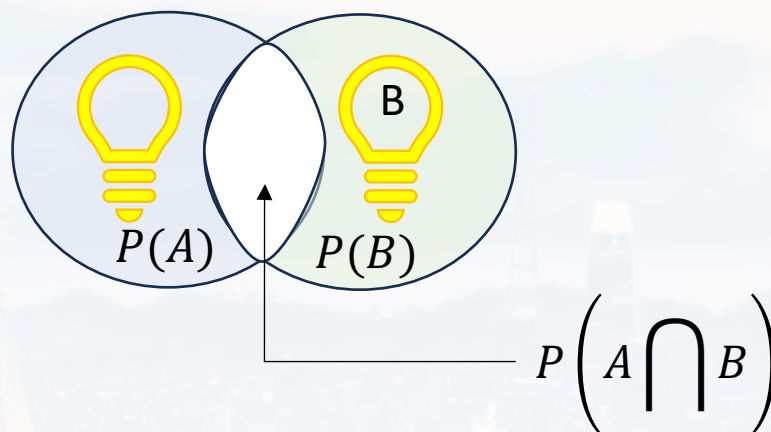
size of **blue** + **green** area: $P(A) + P(B) - P(A \cap B)$



two light bulbs A and B: **EX**clusive **OR**

What is the probability
that A **or** B is turned on?

Venn diagram:



Note: Be careful with Venn diagrams!
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U: “union”

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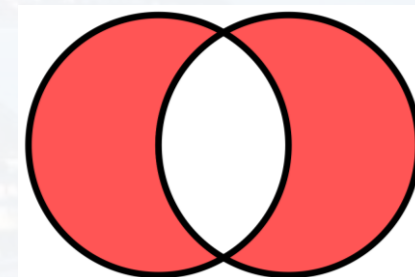
size of **blue** area: $P(A)$

size of **green** area: $P(B)$

size of **blue** + **green** area: $P(A) + P(B) - P(A \cap B)$

EXclusive **OR**: removing the overlapping part → making a “hole”

$$P\left(A \cup B\right) = P(A) + P(B) - 2P\left(A \cap B\right)$$

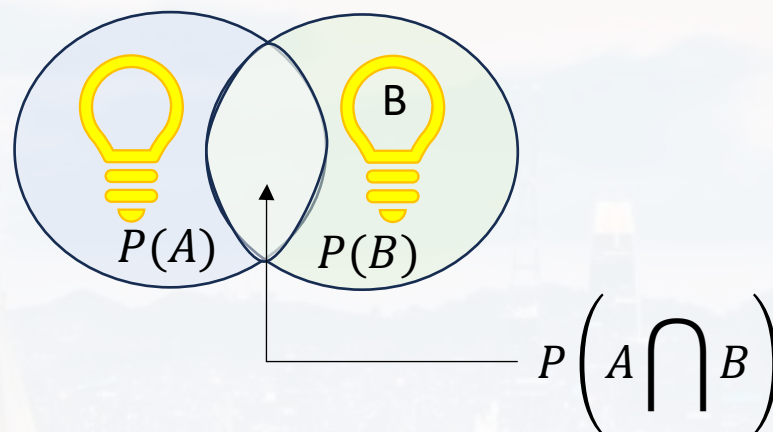




two light bulbs A and B: **I**nclusive **O**R

What is the probability
that A **or** B is turned on?

Venn diagram:



Note: Be careful with Venn diagrams!
They are not a mathematical proof!

$\cup$: “union”

$\cap$: “intersection”

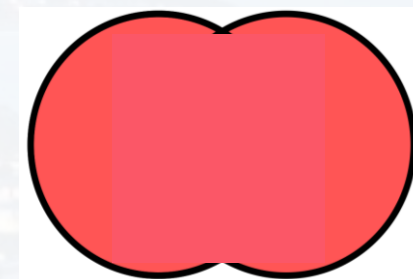
size of **blue** area: $P(A)$

size of **green** area: $P(B)$

size of **blue** + **green** area: $P(A) + P(B) - P(A \cap B)$

Inclusive **O**R: no “hole”

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$





two light bulbs A and B: **IN**clusive **OR**

What is the probability that A **or** B is turned on?



A



B



A



B



$$P(A) + P(B)$$

U: “union”

∩: “intersection”



A



B



A



B



$$P(A)P(B)$$

$$P\left(A \cup B\right) = P(A) + P(B) - P\left(A \cap B\right)$$

$$= P(A) + P(B) - P(A)P(B)$$



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

$P(e_i)$, that an event e_i occurs

“and” equals multiplication!

$$P\left(\bigcap_{i=1}^{\infty} e_i\right) = \prod_{i=1}^{\infty} P(e_i)$$

$\cup$: “union”

$\cap$: “intersection”

inclusion - exclusion principle (here: INclusive OR)

$$P\left(e_4 \cup e_6\right) = P(e_4) + P(e_6) - P(e_4)P(e_6)$$

$$P\left(A \cup B\right) = P(A) + P(B) - P\left(A \cap B\right)$$

$$P\left(A \cup B \cup C\right) = P(A) + P(B) + P(C) + P(A)P(B)P(C) - P(A)P(B) - P(A)P(C) - P(C)P(B)$$



“or” equals addition!

$$P(\cup_{i=1}^{\infty} e_i) = \sum_{i=1}^{\infty} P(e_i)$$

$P(e_i)$, that an event e_i occurs

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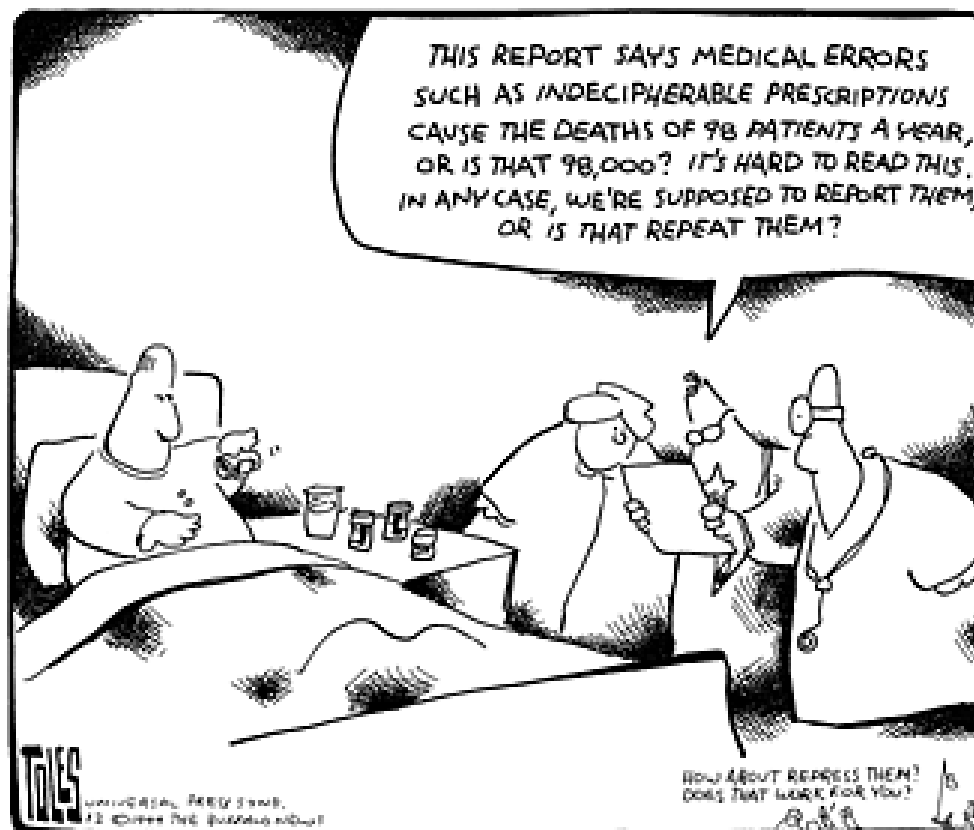
$$P\left(A \cup B\right) = P(A) + P(B) - P\left(A \cap B\right)$$

inclusion - exclusion principle

complement probability for not A, $\bar{A}$:

$$P(\bar{A}) = 1 - P(A)$$

because: $P(e_i) = [0, 1]$ for any e_i



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Outline

- Axioms of Probability

- **Conditional Probabilities and**

Bayes Theorem

- Information and Entropy



$P(A \cap B)$ probability **P** that the events **A** and **B** occur

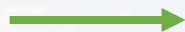
so far: A and B were independent $P(A \cap B) = P(A)P(B) = P(B)P(A)$

now: **conditional probabilities** | “given” or “under the condition”



Thomas Bayes
(1701 - 1761)

$$\begin{aligned} P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A) \end{aligned}$$



$$P(A|B)P(B) = P(B|A)P(A)$$

Bayes Theorem

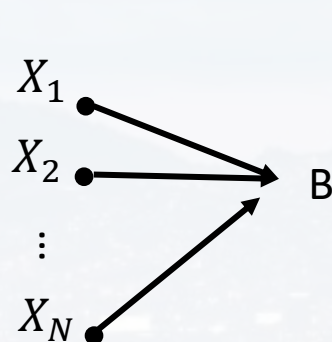
$$\text{posterior } \mathbf{P(A|B)} = \frac{P(B|A)\mathbf{P(A)}}{P(B)} \text{ prior}$$



$$P(A|B)P(B) = P(B|A)P(A)$$

Bayes Theorem

posterior $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$ prior



$$P(B) = \sum_{n=1}^N P(B|X_n)P(X_n)$$

$$P(B) = \int P(B|X)P(X) dX$$

marginalization



Thomas Bayes
(1701 - 1761)

Probability $P(B)$ that I am going to be too late for a meeting:

$$P(B) = P(B|I \text{ forgot that I have a meeting}) P(I \text{ forgot that I have a meeting}) + \\ P(B|I \text{ got sick}) P(I \text{ got sick}) + \\ P(B|BART \text{ was too late}) P(BART \text{ was too late}) + \dots$$



example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Marginalization

$$P(B) = \sum_{n=1}^N P(B|X_n)P(X_n)$$

statement 1: If a person is **diseased**, there is a **95% probability** that the test is **positive**.

statement 2: The **prevalence** for the disease in the average **population** is **0.001%**.

statement 3: **5% of healthy** patients have **a positive result** (aka p-value).

A person takes the test and gets a positive test result. **What is the probability that the person is sick?**

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)} = \frac{\overset{\text{statement 1}}{0.95} P(D)}{P(+)} = \frac{\overset{\text{statement 2}}{0.95 \cdot 0.00001}}{P(+)} = \frac{\overset{\text{marginalization}}{0.95 \cdot 0.00001}}{P(+|D)P(D) + P(+|H)P(H)}$$



example: cancer diagnosis from blood test

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H : health

Bayes Theorem

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$$\begin{aligned}
 P(D|+) &= \frac{P(+|D)P(D)}{P(+)} = \underset{\substack{\uparrow \\ \text{statement 1}}}{0.95} \frac{P(D)}{P(+)} = \underset{\substack{\uparrow \\ \text{statement 2}}}{0.95 \cdot 0.00001} \frac{1}{P(+)} = \frac{0.95 \cdot 0.00001}{P(+|D)P(D) + P(+|H)P(H)} \\
 &\quad \text{marginalization} \\
 &= \frac{0.95 \cdot 0.00001}{P(+|D)P(D) + P(+|H)[1 - P(D)]} \quad \text{complement probability}
 \end{aligned}$$



example: cancer diagnosis from blood test

+ : positive test result
 D : diseased
 H : health

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Marginalization

$$P(B) = \sum_{n=1}^N P(B|X_n)P(X_n)$$

statement 1: If a person is **diseased**, there is a **95% probability** that the test is **positive**.
 statement 2: The **prevalence** for the disease in the average **population** is **0.001%**.
 statement 3: **5% of healthy** patients have **a positive result** (aka p-value).

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|H)[1 - P(D)]}$$

$$= \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}} = \frac{1}{1 + \frac{0.05 [1 - 0.00001]}{0.95 \cdot 0.00001}} = 1/5000$$



example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

statement 1:

sensitivity

$P(D|+) = 95\%$

statement 2:

prior

$P(D) = 0.001\%$

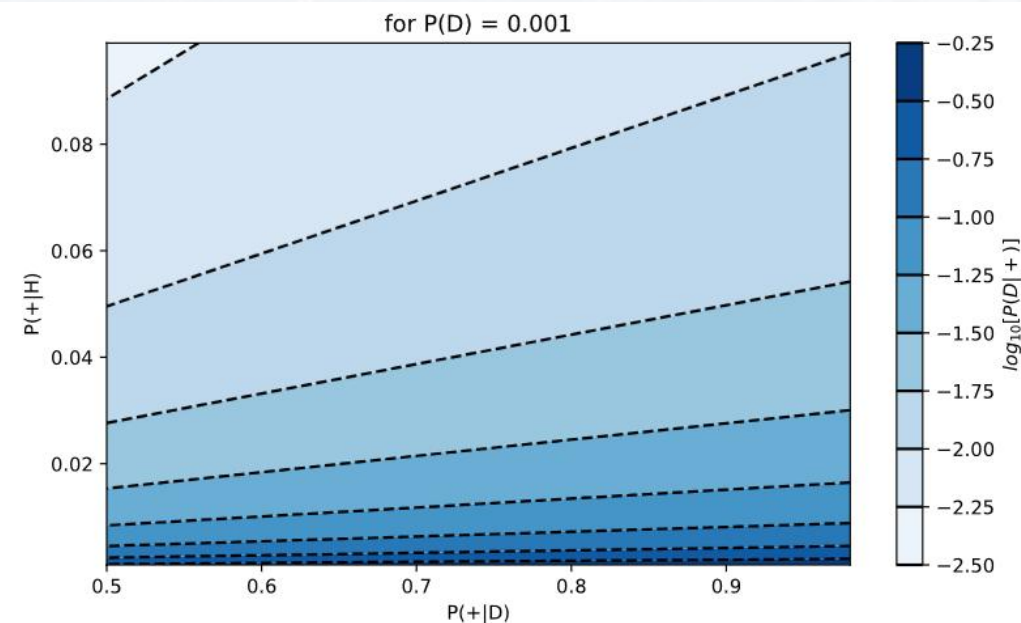
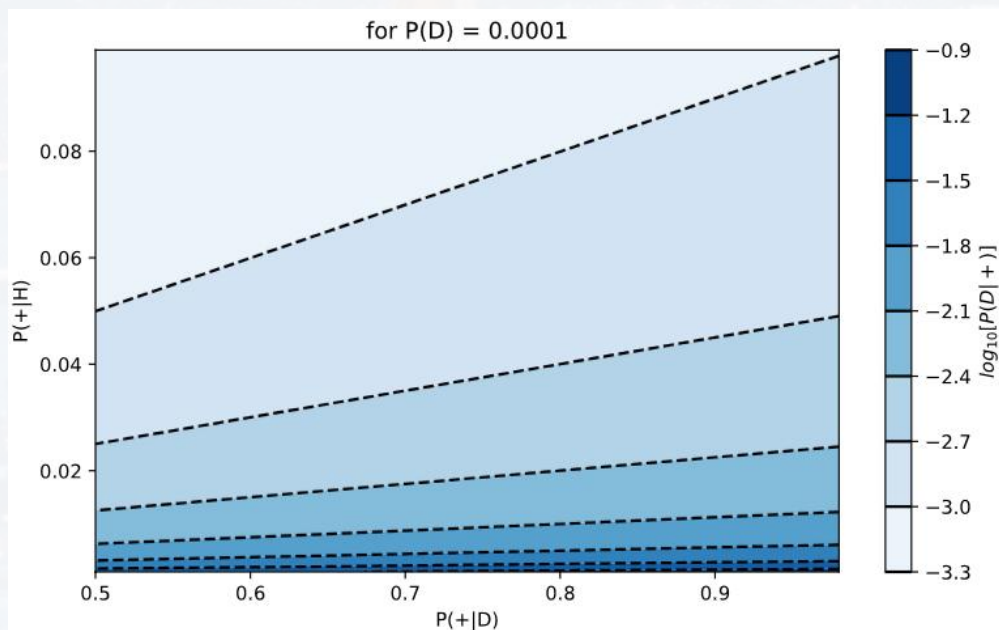
statement 3:

p-value or false positive rate

$P(+|H) = 5\%$

$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

check: `PlotPD_Plus.py`





example: cancer diagnosis from blood test

+ : positive test result
D : diseased
H : health

statement 1:

sensitivity

$P(D|+) = 95\%$

statement 2:

prior

$P(D) = 0.001\%$

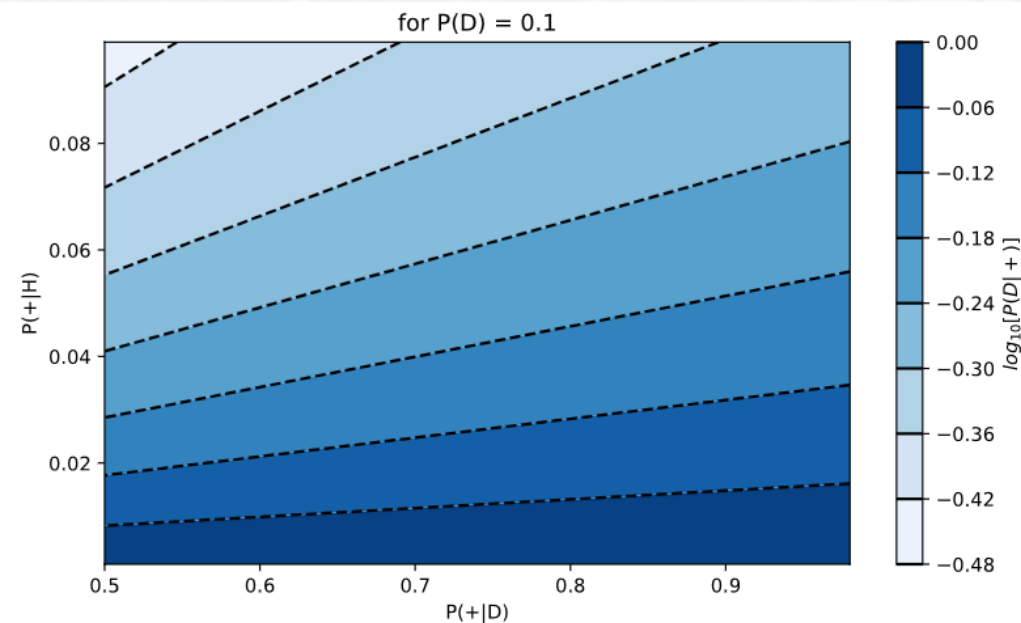
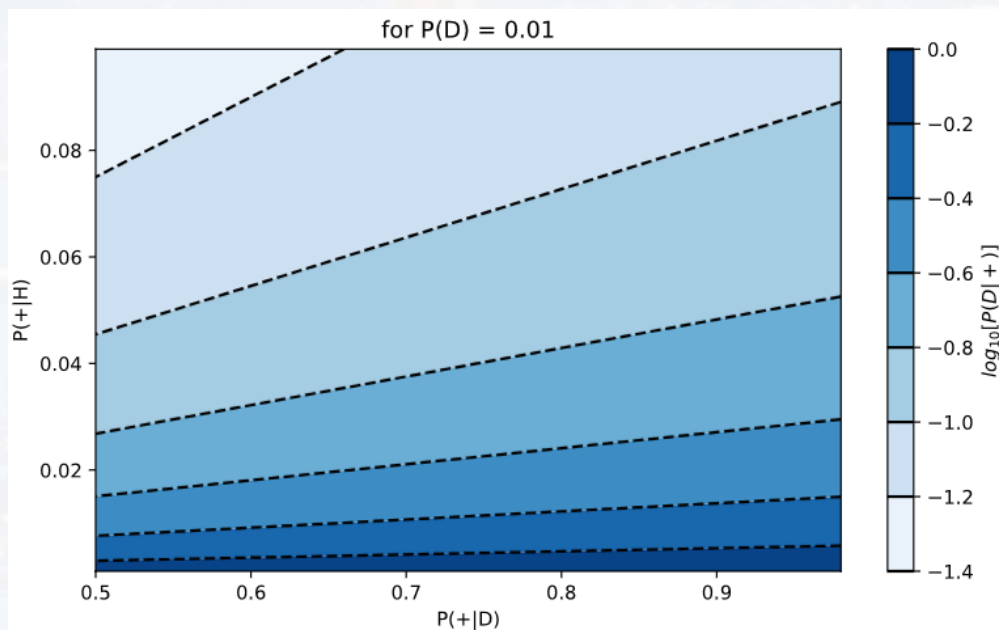
statement 3:

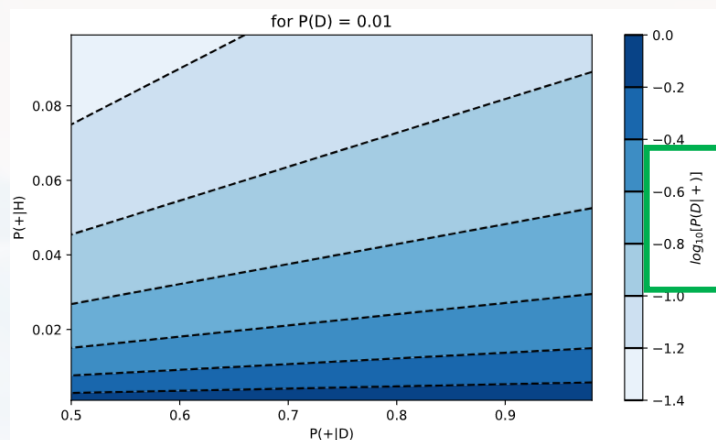
p-value or false positive rate

$P(+|H) = 5\%$

$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

check: `PlotPD_Plus.py`





statement 1:

sensitivity

$P(D|+) = 95\%$

statement 2:

prior

$P(D) = 0.001\%$

statement 3:

p-value or false positive rate

$P(+|H) = 5\%$

odds ratios:

$$\rho_1 = \frac{P(+|H)}{P(+|D)}$$

$$\rho_2 = \frac{1 - P(D)}{P(D)}$$

$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

log odds ratios:

$$r_1 = \log \left[\frac{P(+|H)}{P(+|D)} \right]$$

$$r_2 = \log \left[\frac{1 - P(D)}{P(D)} \right]$$

$$P(D|+) = \frac{1}{1 + e^{r_1} e^{r_2}}$$



log odds ratios: $r_1 = \log \left[\frac{P(+|H)}{P(+|D)} \right]$

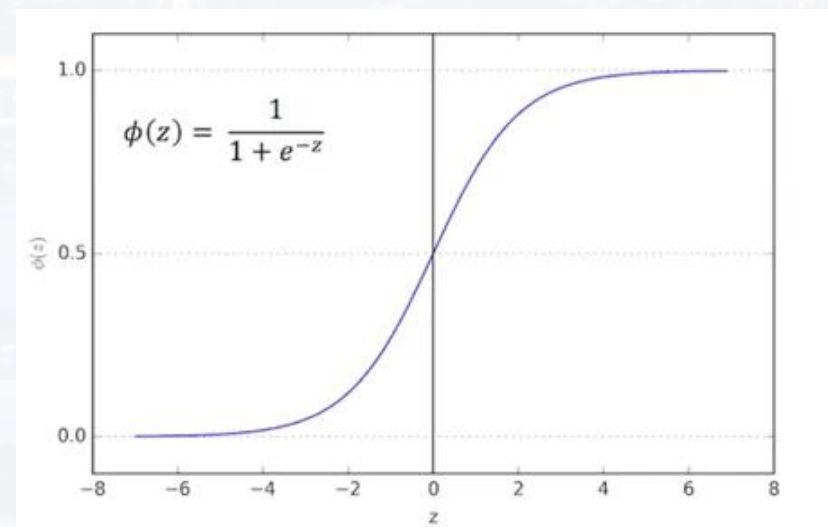
$$r_2 = \log \left[\frac{1 - P(D)}{P(D)} \right]$$

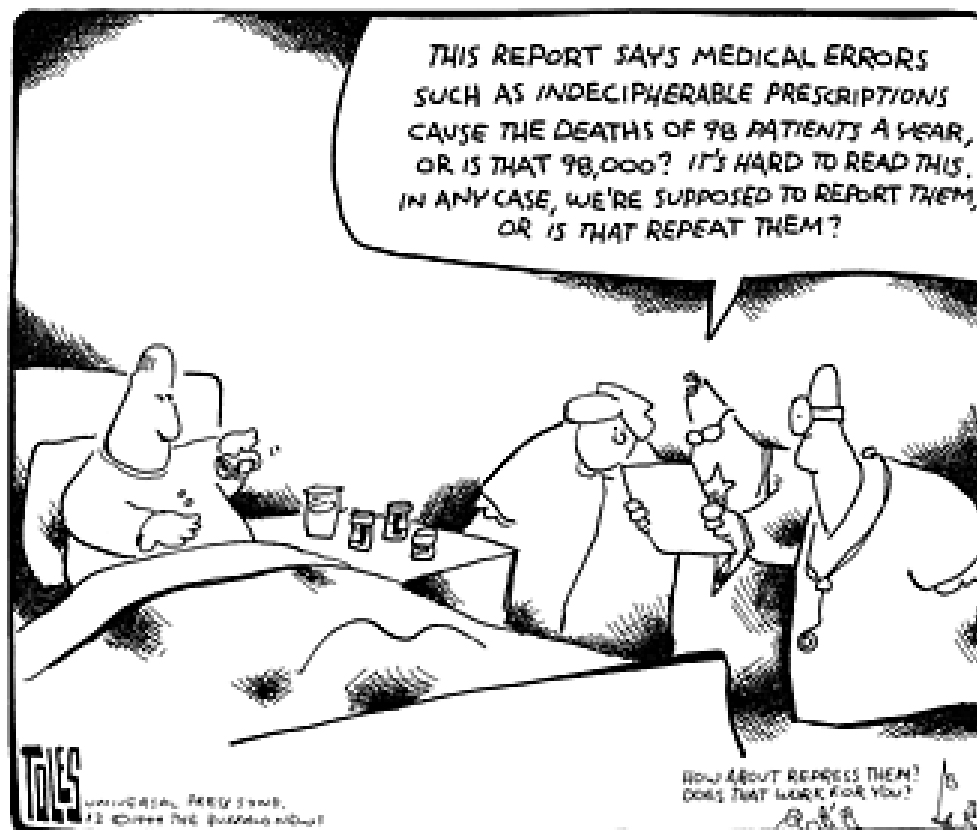
$$P(D|+) = \frac{1}{1 + \frac{P(+|H)[1 - P(D)]}{P(+|D)P(D)}}$$

$$P(D|+) = \frac{1}{1 + e^{r_1}e^{r_2}}$$

logistic (or logit or sigmoid) function

- logistic regression
- transfer function ANN
- bound growth (Verhulst equation)
- binding affinity ligand/receptor





Outline

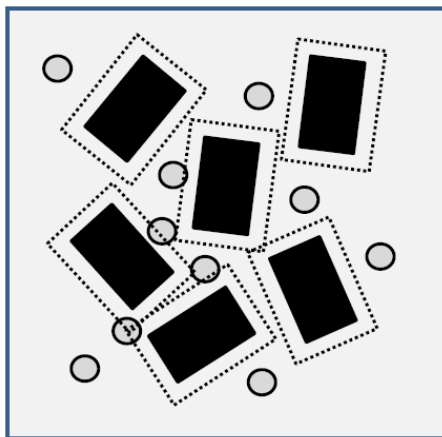
- Axioms of Probability
- Conditional Probabilities and Bayes Theorem
- **Information and Entropy**



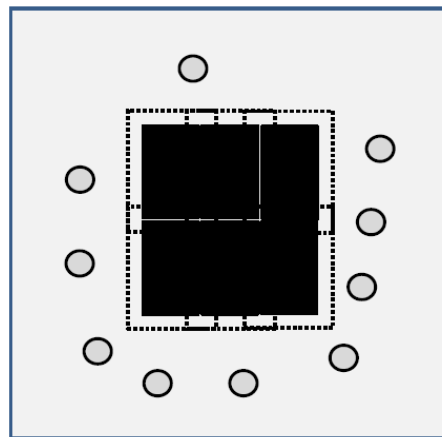
What do you think: in which image is entropy higher?

first of all:

Entropy is **not** a measure of disorder!

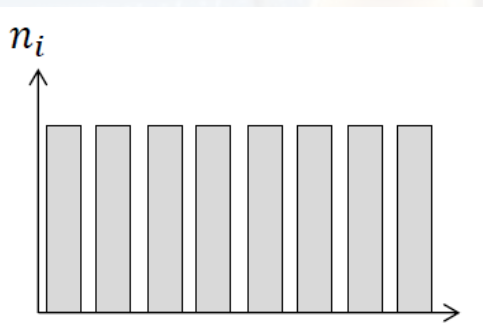


low entropy

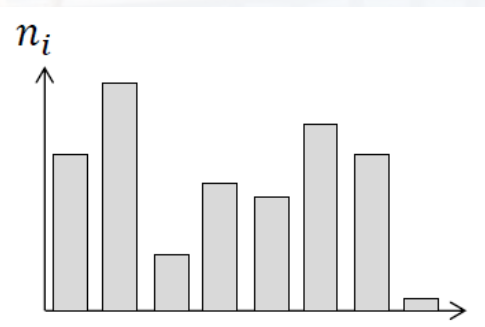


high entropy

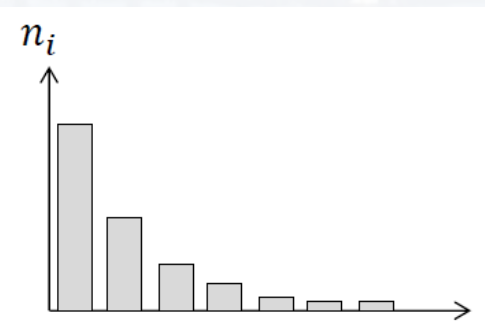
i : states
 n_i : number of particles in state i



highest **possible** entropy



low entropy



high entropy



Often people explain entropy with an ordered vs messy office...



...and then say, that entropy (disorder) grows with time (in closed systems).



first of all:

Entropy is **not** a measure of disorder!

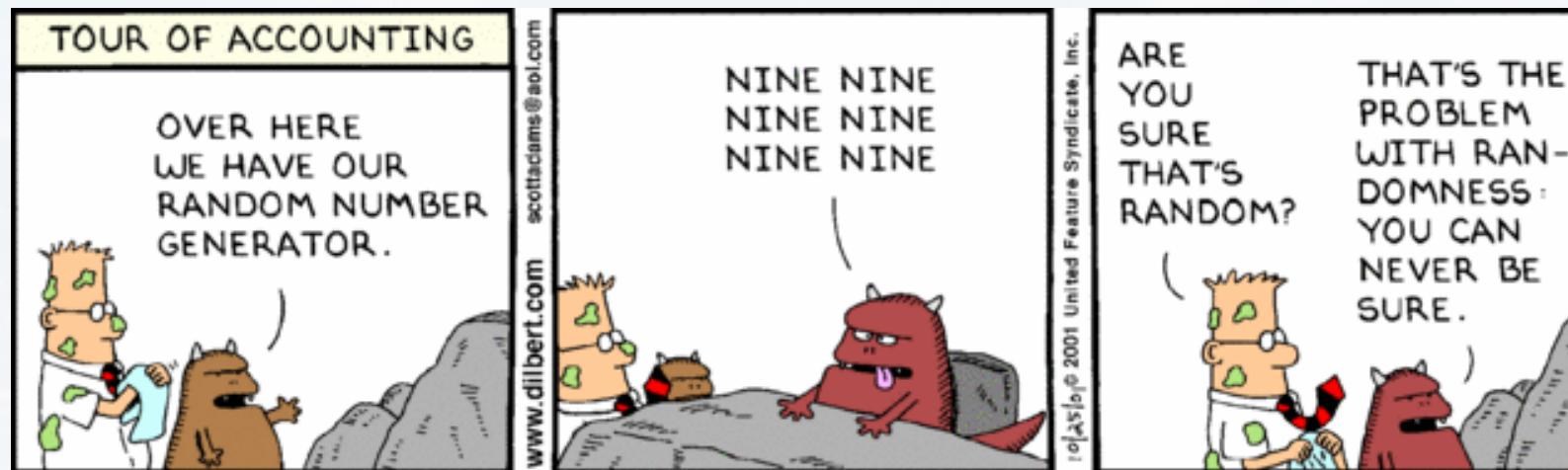
- But how is it possible, that an office can do that, just by itself?
- What if my office just *looks* messy, but I can still pull any file you are asking me for?

order/disorder is not a physical quantity!

Those examples have nothing to do with entropy conceptionally!



actually, the idea of entropy is more like that:



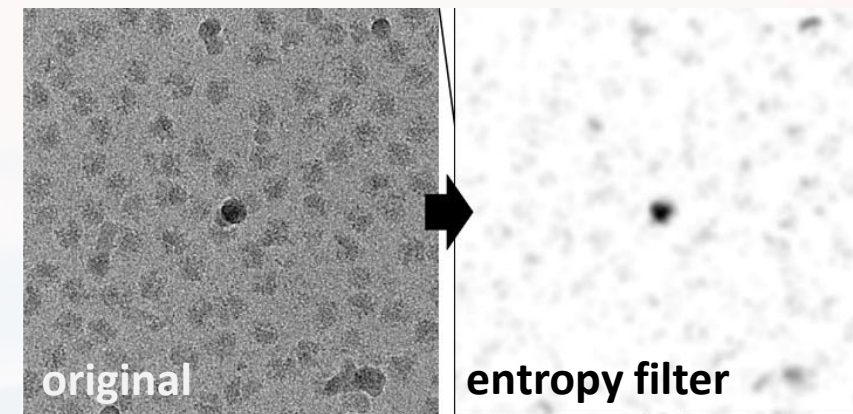


Entropy:

data analysis:

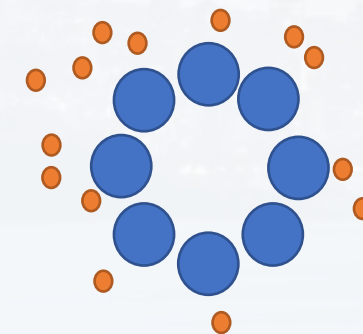
- image processing
- noise reduction
- feature detection

Cryo-EM image of ribosomes



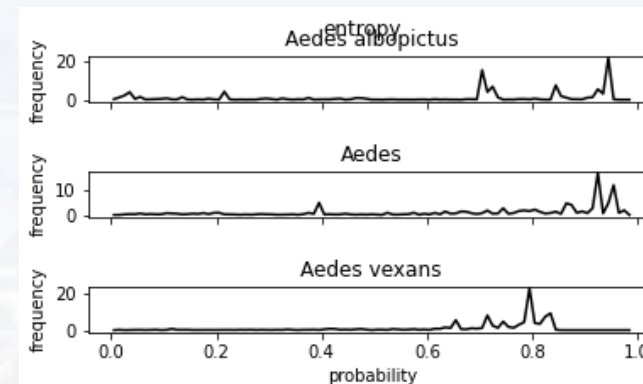
biophysics:

- molecular driving forces
- formation of macromolecules
- “ordering forces”



AI:

- optimization
- cross entropy





Entropy:

statistics/information theory:

- maximum entropy, given constraints

Distribution name	Probability density / mass function	Maximum Entropy constraint	Support
Uniform (discrete)	$f(k) = \frac{1}{b - a + 1}$	None	$\{a, a + 1, \dots, b - 1, b\}$
Uniform (continuous)	$f(x) = \frac{1}{b - a}$	None	$[a, b]$
Bernoulli	$f(k) = p^k (1 - p)^{1-k}$	$\mathbb{E}[K] = p$	$\{0, 1\}$
Geometric	$f(k) = (1 - p)^{k-1} p$	$\mathbb{E}[K] = \frac{1}{p}$	$\mathbb{N} \setminus \{0\} = \{1, 2, 3, \dots\}$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$\mathbb{E}[X] = \frac{1}{\lambda}$	$[0, \infty)$
Laplace	$f(x) = \frac{1}{2b} \exp\left(-\frac{ x - \mu }{b}\right)$	$\mathbb{E}[X - \mu] = b$	$(-\infty, \infty)$
Asymmetric Laplace	$f(x) = \frac{\lambda \exp(-(x - m) \lambda s \kappa^s)}{(\kappa + \frac{1}{\kappa})}$ where $s \equiv \text{sgn}(x - m)$	$\mathbb{E}[(X - m) s \kappa^s] = \frac{1}{\lambda}$	$(-\infty, \infty)$
Pareto	$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$	$\mathbb{E}[\ln X] = \frac{1}{\alpha} + \ln(x_m)$	$[x_m, \infty)$
Normal	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$	$\mathbb{E}[X] = \mu,$ $\mathbb{E}[X^2] = \sigma^2 + \mu^2$	$(-\infty, \infty)$
Gamma	$f(x) = \frac{x^{k-1} \exp\left(-\frac{x}{\theta}\right)}{\theta^k \Gamma(k)}$	$\mathbb{E}[X] = k \theta,$ $\mathbb{E}[\ln X] = \psi(k) + \ln \theta$	$[0, \infty)$



in this section:

entropy S is a **measure of information** we have **about a system**

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

problem:

I want to rewatch a series on Netflix.

Where have I left off, i. e. which episodes have I watched already ?





entropy S is a **measure of information** we have **about a system**

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I want to rewatch a series on Netflix. Where have I left off, i. e. which episodes have I watched already?

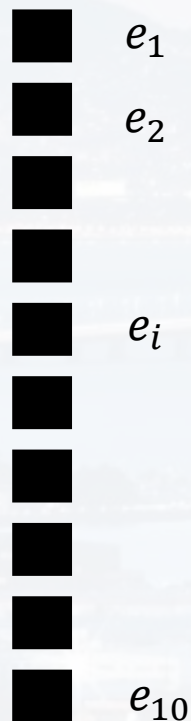
$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

p_i : probability that I have watched episode e_i

case 1):

no idea, no information $\rightarrow$ all $p_i = 0.5$

list of episodes





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list of episodes



e_1



e_2



e_i



e_{10}

$$S = - \sum_{i=1}^{10} \frac{1}{2} \ln\left(\frac{1}{2}\right) = \sum_{i=1}^{10} \frac{1}{2} \ln(2) = 5 \ln(2) \approx \mathbf{3.46}$$



entropy S is a **measure of information** we have **about a system**

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I want to rewatch a series on Netflix. Where have I left off, i. e. which episodes have I watched already?

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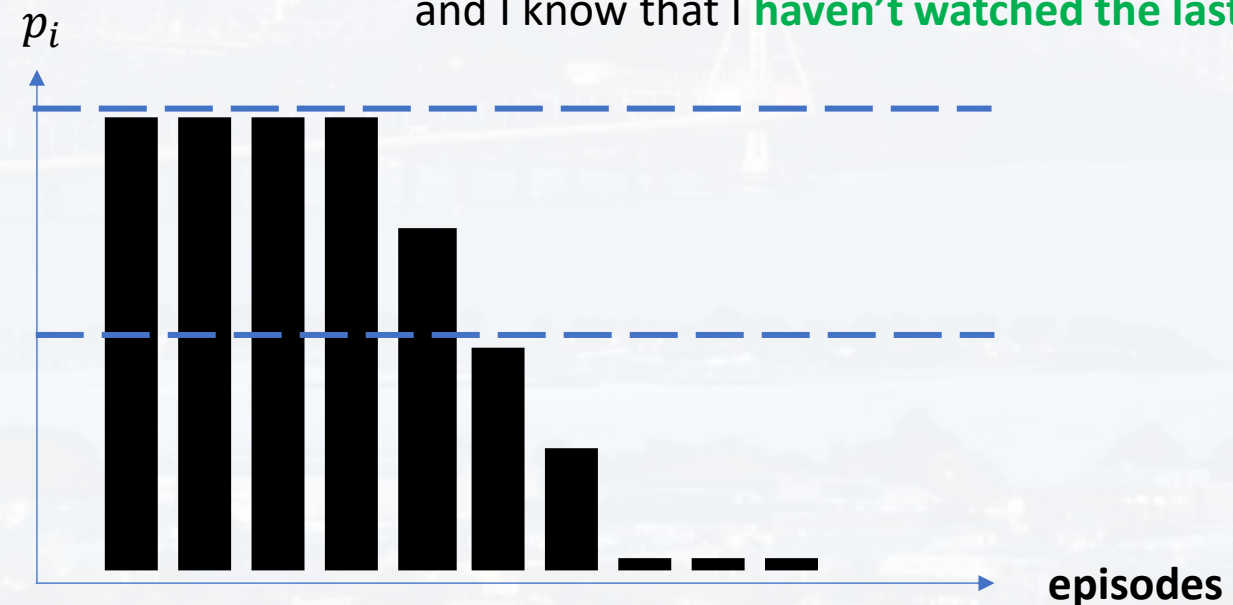
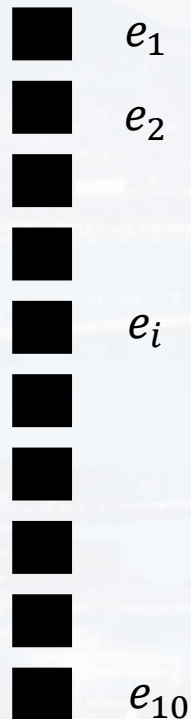
case 1):

no idea, no information $\rightarrow$ all $p_i = 0.5$ $S \approx 3.46$

case 2):

usually, I remember that I have **watched some of the first episodes**, I am **not sure about 2 or 3 episodes** and I know that I **haven't watched the last episodes**

list of episodes





entropy S is a **measure of information** we have **about a system**

problem:

I want to rewatch a series on Netflix. Where have I left off, i. e. which episodes have I watched already?

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usually, I remember that I have **watched some of the first episodes**, I am **not sure about 2 or 3 episodes** and I know that I **haven't watched the last episodes**

list of episodes



$$S = - \sum_{i=1}^4 1 \ln(1) - 0.75 \ln(0.75) - 0.5 \ln(0.5)$$

$$- 0.25 \ln(0.25) - \sum_{i=1}^3 0 \ln(0) \approx 0.91$$



entropy S is a **measure of information** we have **about a system**

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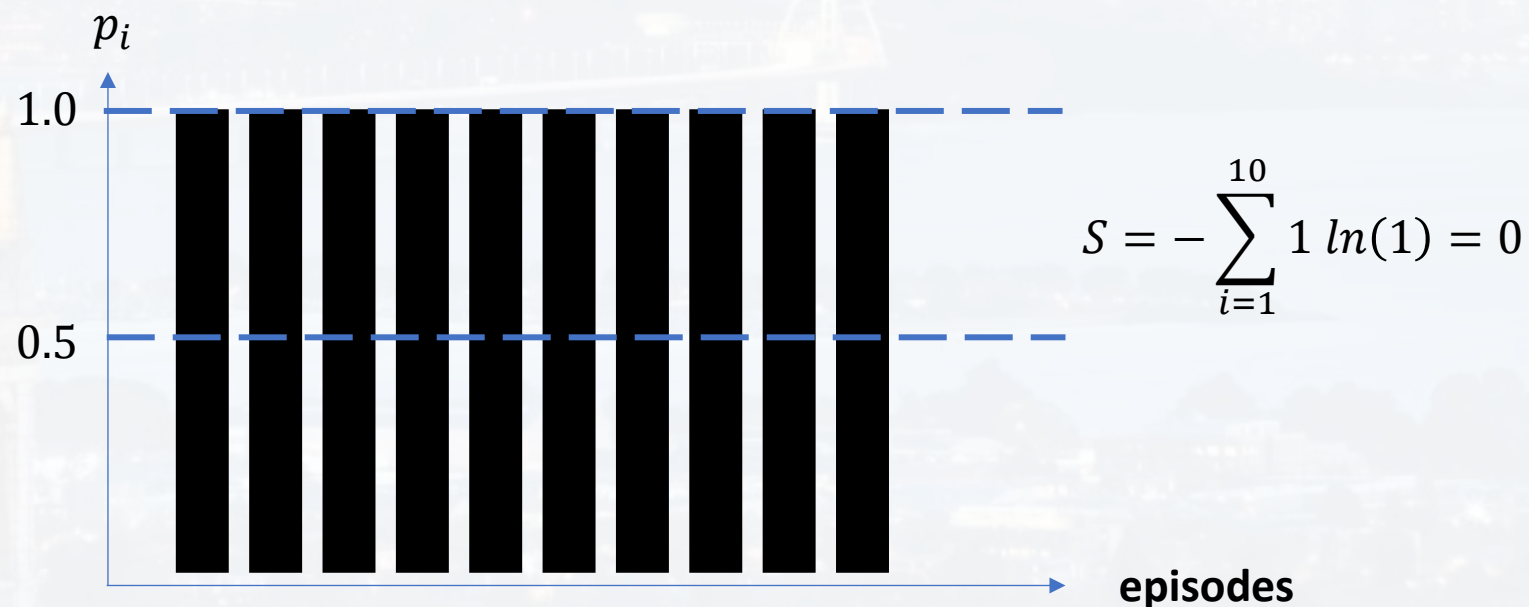
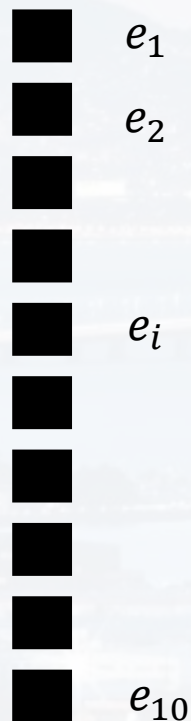
p_i : probability that I have watched episode e_i

case 1): no idea, no information $\rightarrow$ all $p_i = 0.5$ $S \approx 3.46$

case 2): some information $S \approx 0.91$

case 3): **all** information

list of episodes





entropy S is a **measure of information** we have **about a system**

problem:

I want to rewatch a series on Netflix. Where have I left off, i. e. which episodes have I watched already?

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

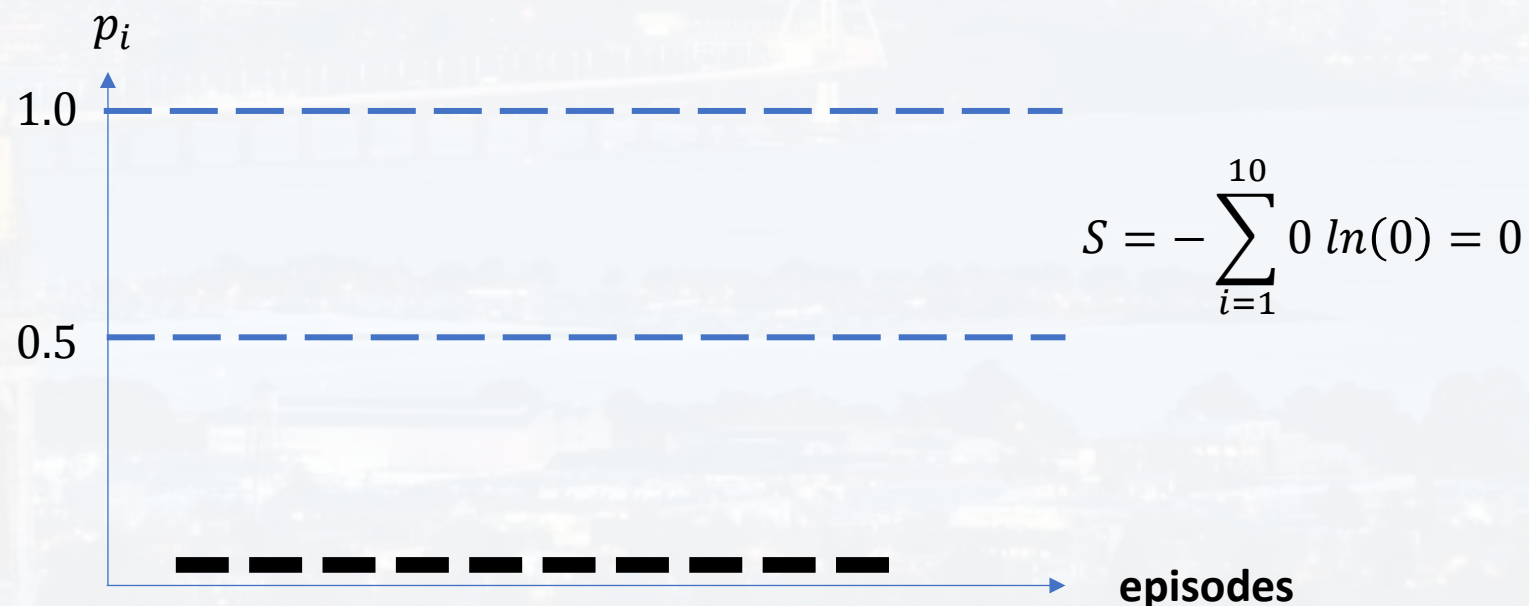
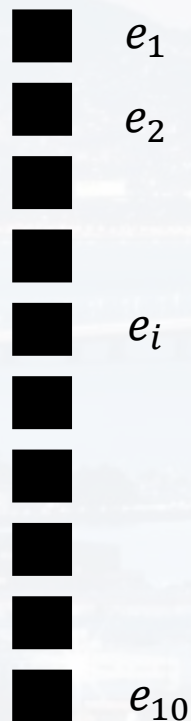
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case 3): **all** information

list of episodes





entropy S is a **measure of information** we have **about a system**

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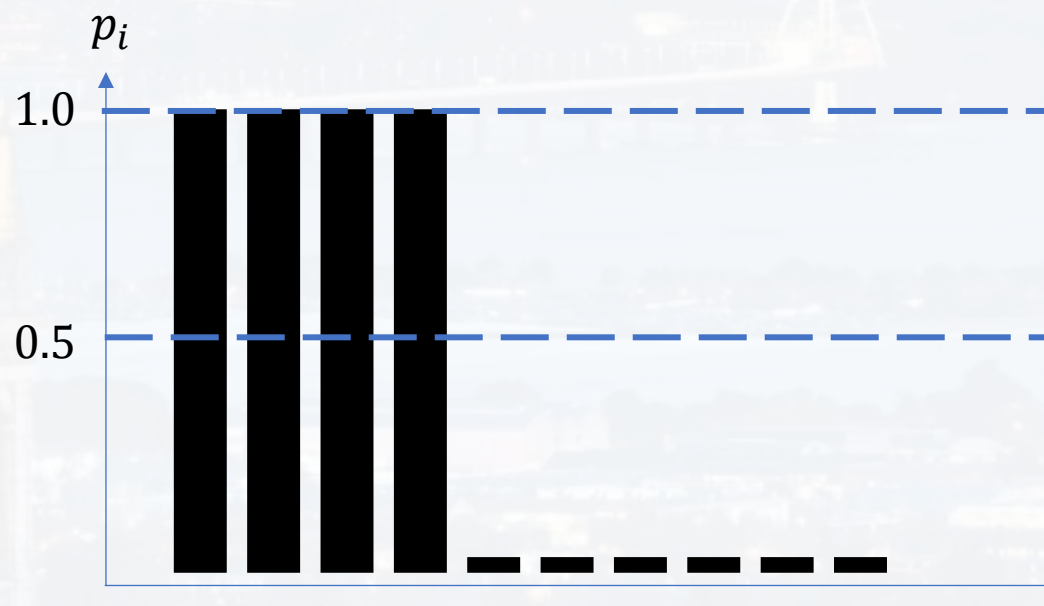
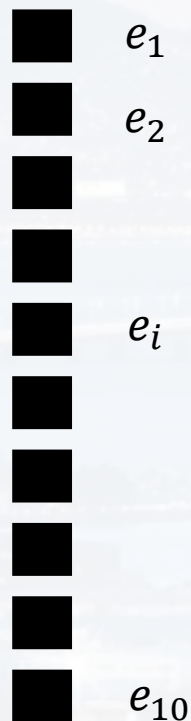
p_i : probability that I have watched episode e_i

case 1): no idea, no information $\rightarrow$ all $p_i = 0.5$ $S \approx 1.73$

case 2): some information $S \approx 0.91$

case 3): **all** information

list of episodes



$$S = - \sum_{i=1}^6 0 \ln(0) + \sum_{i=1}^4 1 \ln(1) = 0$$

episodes



entropy S is a **measure of information** we have **about a system**

problem:

I want to rewatch a series on Netflix. Where have I left off, i. e. which episodes have I watched already?

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

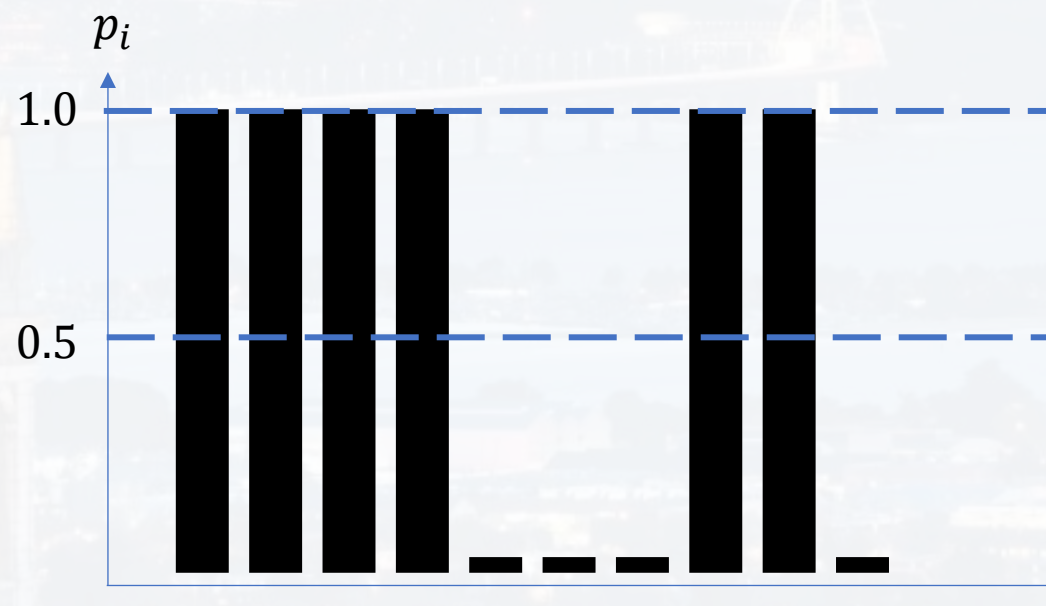
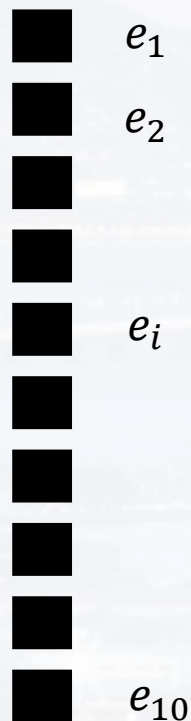
p_i : probability that I have watched episode e_i

case 1): no idea, no information $\rightarrow$ all $p_i = 0.5$ $S \approx 1.73$

case 2): some information $S \approx 0.91$

case 3): **all** information

list of episodes



As long as the p_i are zero or one (i. e. I know **exactly** if I have watched the episode) $\rightarrow$ entropy = 0



entropy S is a **measure of information** we have **about a system**

problem:

I want to rewatch a series on Netflix. Where have I left off, i. e. which episodes have I watched already?

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

p_i : probability that I have watched episode e_i

list of episodes



e_1



e_2



e_i



e_{10}

case 1): no idea, no information $\rightarrow$ all $p_i = 0.5$ $S \approx 1.73$

case 2): some information $S \approx 0.91$

case 3): **all** information $S \approx 0.00$

The lower the entropy, the more information!

If $p_i = 0.5 = \bar{p}_i$, maximum entropy (uniform distribution)!



entropy S is a **measure of information** we have **about a system**

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

two states $\uparrow$ or $\downarrow$

and three entities $\rightarrow$ system

possible states of the system

all three up

two up

one up

all three down

$\uparrow\uparrow\uparrow$

$\uparrow\uparrow\downarrow$

$\uparrow\downarrow\downarrow$

$\downarrow\downarrow\downarrow$

$\uparrow\downarrow\uparrow$

$\downarrow\uparrow\downarrow$

$\downarrow\uparrow\uparrow$

$\downarrow\downarrow\uparrow$

eight possible states

no idea (before the experiment)

$$S = - \sum_{i=1}^8 \frac{1}{8} \ln\left(\frac{1}{8}\right) = \sum_{i=1}^8 \frac{1}{8} \ln(8) = \ln(8) \approx \mathbf{2.08}$$



entropy S is a **measure of information** we have **about a system**

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

two states $\uparrow$ or $\downarrow$

and three entities $\rightarrow$ system

possible states of the system

all three up

two up

one up

all three down

$\uparrow\uparrow\uparrow$

$\uparrow\uparrow\downarrow$

$\uparrow\downarrow\downarrow$

~~$\downarrow\downarrow\downarrow$~~

$\uparrow\downarrow\uparrow$

$\downarrow\uparrow\downarrow$

$\downarrow\uparrow\uparrow$

$\downarrow\downarrow\uparrow$

one measurement
 $\rightarrow$ at least one arrow up

seven possible states

$$S = \sum_{i=1}^7 \frac{1}{7} \ln(7) + 0 \ln(0) = \ln(7) \approx \mathbf{1.95}$$



entropy S is a **measure of information** we have **about a system**

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

two states $\uparrow$ or $\downarrow$

and three entities $\rightarrow$ system

possible states of the system

all three up

two up

one up

all three down

$\uparrow\uparrow\uparrow$	$\uparrow\uparrow\downarrow$	$\uparrow\downarrow\downarrow$	$\downarrow\downarrow\downarrow$
	$\uparrow\downarrow\uparrow$	$\downarrow\uparrow\downarrow$	
	$\downarrow\uparrow\uparrow$	$\downarrow\downarrow\uparrow$	

second measurement
 $\rightarrow$ another arrow up

four possible states

$$S = \sum_{i=1}^4 \frac{1}{4} \ln(4) = \ln(4) \approx \mathbf{1.39}$$



entropy S is a **measure of information** we have **about a system**

$$S = - \sum_{i=1}^I p_i \ln(p_i)$$

two states $\uparrow$ or $\downarrow$

and three entities $\rightarrow$ system

possible states of the system

all three up

~~$\uparrow\uparrow\uparrow$~~

two up

$\uparrow\uparrow\downarrow$

$\uparrow\downarrow\uparrow$

$\downarrow\uparrow\uparrow$

one up

~~$\uparrow\downarrow\downarrow$~~

~~$\downarrow\uparrow\downarrow$~~

~~$\downarrow\downarrow\uparrow$~~

all three down

~~$\downarrow\downarrow\downarrow$~~

third measurement
 $\rightarrow$ one arrow down

three possible states

$$S = \ln(3) \approx \mathbf{1.10}$$

The lower the entropy, the more information!



Thank you very much for your attention!

